

Predicting elbow varus torque in pitchers a model has never seen

How much of a pitcher's elbow load can be predicted from body size, velocity and mechanics, and how much does leaky validation overstate it?

Andres Perez, M.S. Kinesiology
(Biomechanics)

Python · statsmodels · scikit-learn

Code: github.com/Andresperez397/obp-elbow-torque

DATA

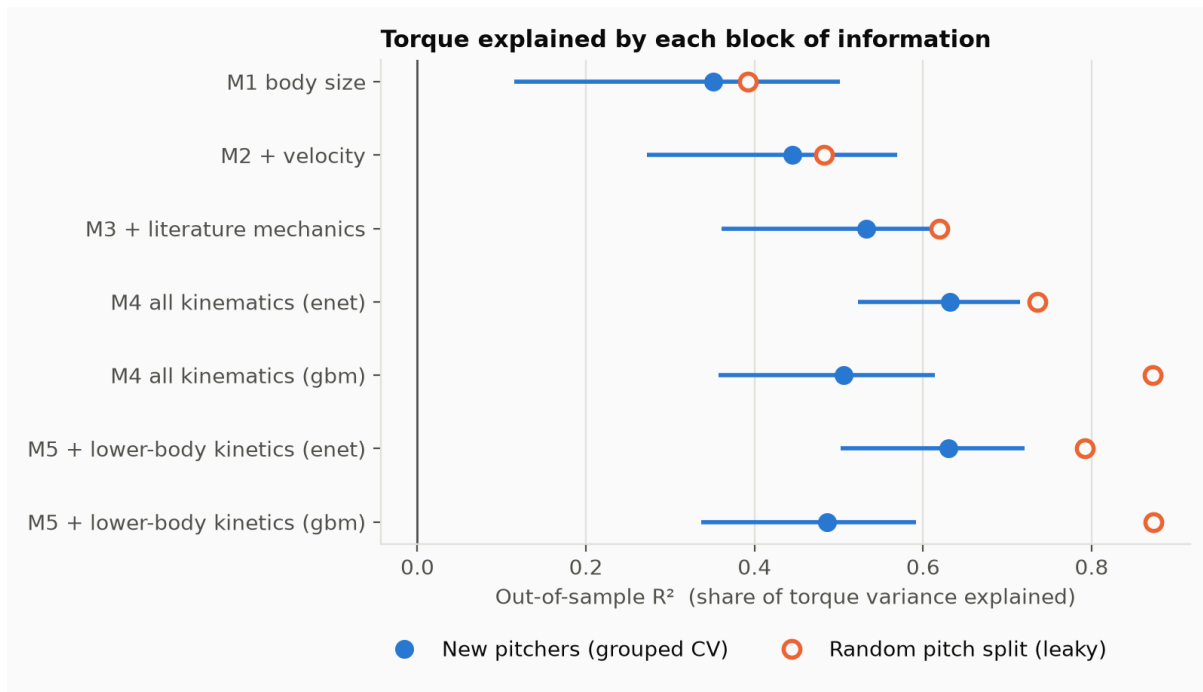
411 fastballs from 100 pitchers in the OpenBiomechanics Project (Driveline Baseball): marker-based motion capture (360 Hz) and force plates, processed by inverse dynamics. The pitchers are mostly college level, with one session each. The outcome is peak elbow varus torque (Nm).

APPROACH

- **Pre-registered:** the questions, models and decision rules were committed to git before any model was fit.
- **No leakage:** predictors from the same arm inverse-dynamics solution were excluded.
- **Grouped validation:** by pitcher, 10 folds $\times$ 20 repeats; all preprocessing done inside folds.
- **Uncertainty:** pitcher-level bootstrap CIs; mixed models for the variance split and velocity effects; 15 unit tests, including leakage checks.

FINDINGS

- **Torque is a pitcher trait:** 95% of the variance lies between pitchers (ICC 0.95).
- **Mechanics add real information:** on unseen pitchers, explained variance rises from R^2 0.45 (body size + velocity) to 0.63 (CI 0.52–0.72) with the full kinematic set.
- **Lower-body kinetics and ground reaction forces add nothing** beyond kinematics.
- **Leaky validation misleads:** random pitch splits made boosted trees look best (R^2 0.87). On new pitchers they reach 0.51, below the linear model.
- **Velocity cost:** about 1.3 Nm per mph, both between and within pitchers.

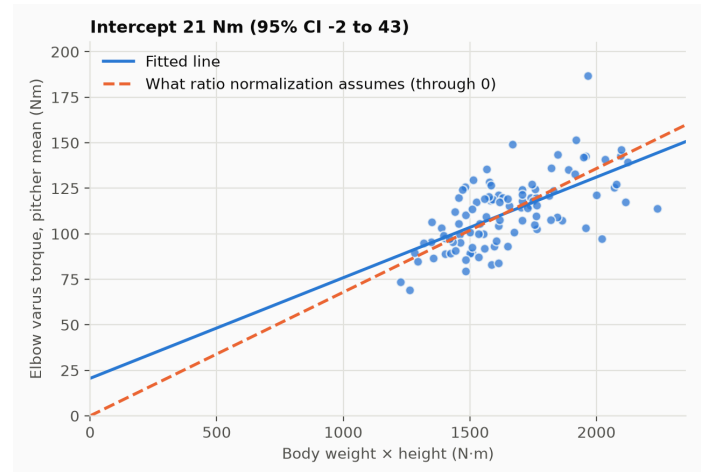
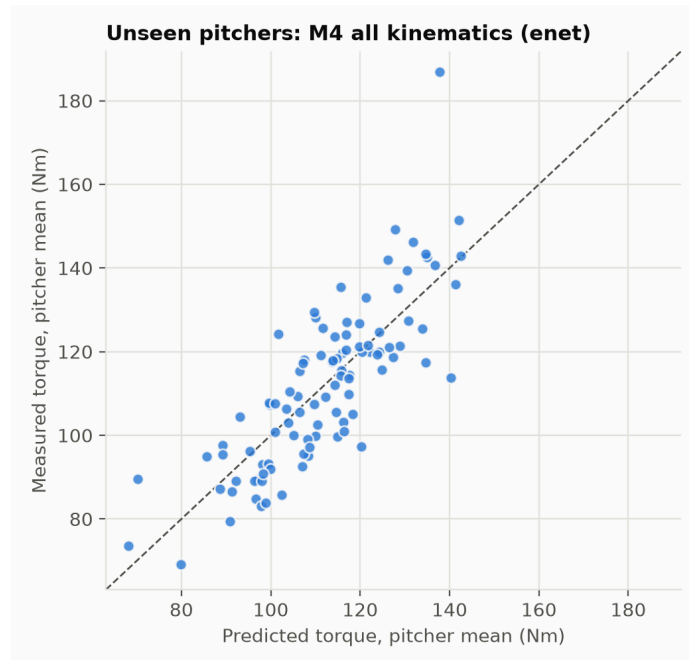


Out-of-sample R^2 for each information block. Filled: validated on held-out pitchers, with 95% CI. Hollow: the same model under a random pitch split.

MODEL RESULTS (HELD-OUT PITCHERS)

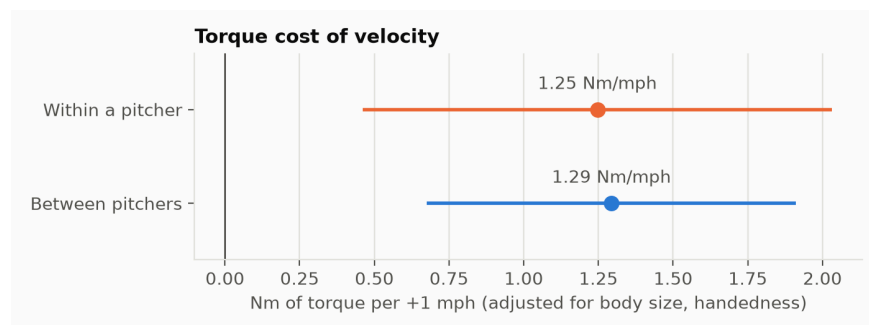
Model	R ²	95% CI	RMSE (Nm)	R ² , leaky split
Body size + handedness	0.35	0.12–0.50	16.0	0.39
+ velocity	0.45	0.27–0.57	14.8	0.48
+ 10 literature mechanics	0.53	0.36–0.63	13.6	0.62
All kinematics, elastic net	0.63	0.52–0.72	12.0	0.74
All kinematics, boosted trees	0.51	0.36–0.61	14.0	0.87
+ lower-body kinetics, elastic net	0.63	0.50–0.72	12.1	0.79

R² and RMSE: averages over 20 repeats of 10-fold CV grouped by pitcher. CIs: pitcher bootstrap (2,000 resamples) of the first repeat.



Ratio normalization assumes a line through zero. The intercept CI (–2 to 43 Nm) includes zero, so normalization is defensible here.

Pitcher-mean torque against the prediction for pitchers held out of training.



LIMITATIONS

The sample is 100 mostly college pitchers tested in one lab, throwing fastballs, so results may not transfer to MLB or in-game settings. Torque is an inverse-dynamics estimate whose level scales with body mass by construction. With 100 pitchers, the intervals for R² gains span about ± 0.1 , so smaller differences are not findings.